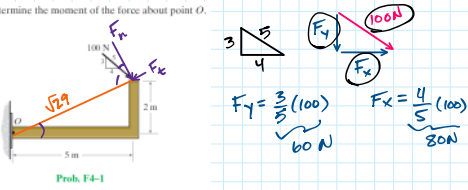


F4-1. Determine the moment of the force about point O.

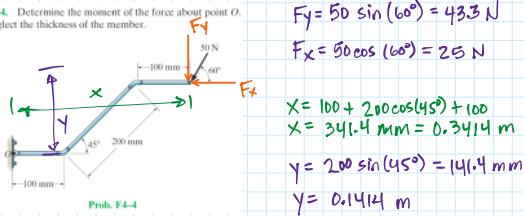


$$M_O = -(2\text{ m})(80\text{ N}) - (5\text{ m})(60\text{ N})$$

$$M_O = -460\text{ N}\cdot\text{m}$$

$$\boxed{+M_O = 460\text{ N}\cdot\text{m} \curvearrowright}$$

F4-4. Determine the moment of the force about point O. Neglect the thickness of the member.



$$F_y = 50 \sin(60^\circ) = 43.3\text{ N}$$

$$F_x = 50 \cos(60^\circ) = 25\text{ N}$$

$$X = 100 + 200 \cos(45^\circ) + 100$$

$$X = 341.4\text{ mm} = 0.3414\text{ m}$$

$$Y = 200 \sin(45^\circ) = 141.4\text{ mm}$$

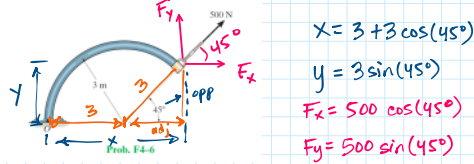
$$Y = 0.1414\text{ m}$$

$$\curvearrowright M_O = -X F_y + Y F_x = -(0.3414)(43.3) + (0.1414)(25) =$$

$$\curvearrowright M_O = -14.78 + 3.54 = -11.22\text{ N}\cdot\text{m}$$

$$\boxed{\curvearrowright M_O = -11.22\text{ N}\cdot\text{m} \quad \text{or} \quad 11.22\text{ N}\cdot\text{m} \curvearrowright}$$

F4-6. Determine the moment of the force about point O.



$$X = 3 + 3 \cos(45^\circ)$$

$$Y = 3 \sin(45^\circ)$$

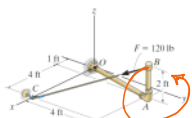
$$F_x = 500 \cos(45^\circ)$$

$$F_y = 500 \sin(45^\circ)$$

$$\curvearrowright M_O = -Y F_x + X F_y = 1,060\text{ N}\cdot\text{m}$$

$$\boxed{M_O = 1060\text{ N}\cdot\text{m} \text{ ccw}}$$

F4-11. Determine the moment of force F about point O. Express the result as a Cartesian vector.



$$\vec{F} = \underline{\hspace{1cm}} \hat{i} + \underline{\hspace{1cm}} \hat{j} + \underline{\hspace{1cm}} \hat{k}$$

$$F = 120\text{ lb}$$

Solve for unit vector in direction of BC.

$$\frac{\vec{BC}}{|\vec{BC}|}$$

$$\vec{OB} + \vec{BC} = \vec{OC}$$

$$\vec{OB} = \langle 2, 4, 2 \rangle$$

$$\vec{OC} = \langle 5, 0, 0 \rangle$$

$$\rightarrow \vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = \langle 3, -4, -2 \rangle$$

$$|\vec{BC}| = \sqrt{3^2 + (-4)^2 + (-2)^2}$$

$$|\vec{BC}| = \sqrt{29} = 5.385$$

$$\frac{\vec{BC}}{|\vec{BC}|} = \left\langle \frac{3}{5.385}, -\frac{4}{5.385}, -\frac{2}{5.385} \right\rangle$$

$$\vec{F} = 120 \cdot \left\langle \frac{3}{5.385}, -\frac{4}{5.385}, -\frac{2}{5.385} \right\rangle = \langle 67.2, -89.6, -44.8 \rangle$$

$$M_O = \vec{OB} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 4 & 2 \\ 67.2 & -89.6 & -44.8 \end{vmatrix} = \begin{vmatrix} -160 & 160 \\ 4(-40) & -2(-80) \\ -40 & -160 \end{vmatrix} \begin{vmatrix} \hat{i} \\ \hat{j} \\ \hat{k} \end{vmatrix}$$

$$M_0 = 0\hat{i} + 200\hat{j} - 400\hat{k} \quad \text{lb-ft}$$

Diagram of a beam with the following forces and dimensions:

- Support at A: 300 N (upward)
- Force at 3 m: 400 N (downward)
- Force at 5 m: 400 N (upward)
- Force at 7 m: 200 N (upward)
- Force at 7 m: 200 N (downward)
- Dimensions: 3 m, 2 m, 0.2 m

Prob. E4-19

$$+\circlearrowleft \sum M_B = -740 \text{ N}\cdot\text{m}$$

$$+ \sum M_A = -740 \text{ N}\cdot\text{m}$$

68.7°
A
 $M_A = 470 \text{ N}\cdot\text{m}$

$$F_R = 107.7 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{100}{40} \right)$$

$$\theta = 68.2^\circ$$

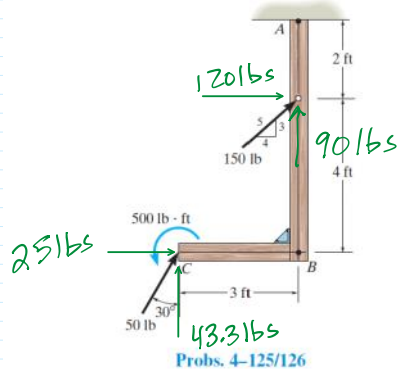
$$\uparrow \sum M_A = (-30\text{N})(3\text{m}) - (30\text{N})(6\text{m}) - 200\text{N}\cdot\text{m}$$

$$+\circlearrowleft M_A = -470 \text{ N}\cdot\text{m}$$

$$\text{or } M_A = 4170 \text{ N}\cdot\text{m} \text{ CW}$$

4-125. Replace the force and couple system acting on the frame by an equivalent resultant force and specify where the resultant's line of action intersects member AB, measured from A.

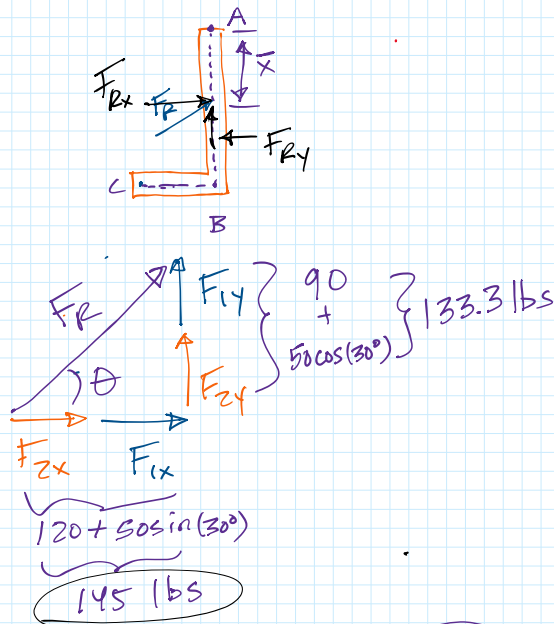
4-126. Replace the force and couple system acting on the frame by an equivalent resultant force and specify where the resultant's line of action intersects member BC, measured from B.



Soln 4-125

$$F_{2x} = 50 \sin(30^\circ)$$

$$F_{2y} = 50 \cos(30^\circ)$$



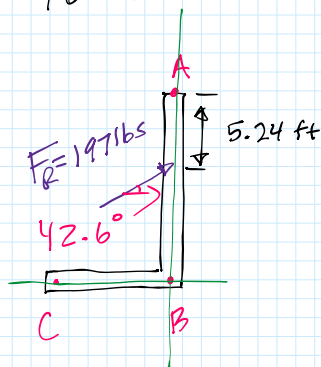
$$F_R = \sqrt{145^2 + 133.3^2} = F_R = 197 \text{ lbs}$$

$$\theta = \tan^{-1}\left(\frac{133.3}{145}\right) = 42.6^\circ$$

$$+\circlearrowleft \sum M_A = 25(6) - 43.3(3) + 120(2) + 500 = 760.1 \text{ lbs}\cdot\text{ft}$$

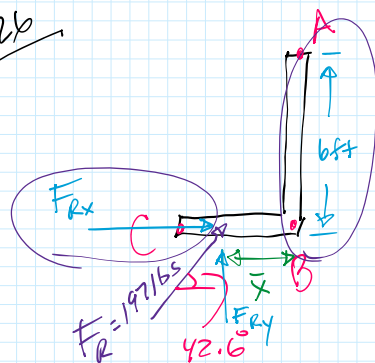
$$M_A = F_{Rx} \cdot \bar{x}$$

$$760.1 = 145 \cdot \bar{x}$$



$$\bar{x} = 5.24 \text{ ft}$$

4-126



$$M_A = 6 \cdot F_{Rx} - F_{Ry} \bar{x}$$

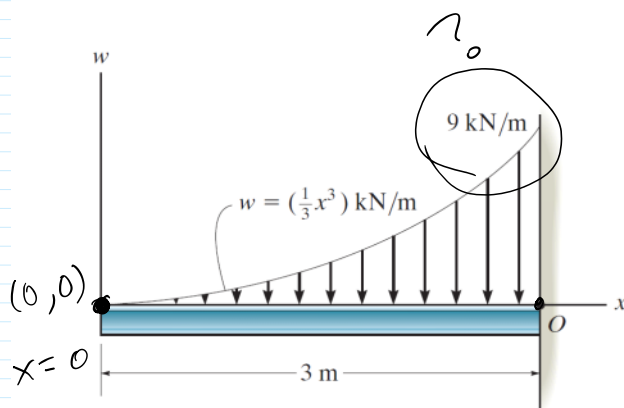
$$760.1 = 6(145) - 133.3 \bar{x}$$

$$760.1 = 870 - 133.3 \bar{x}$$

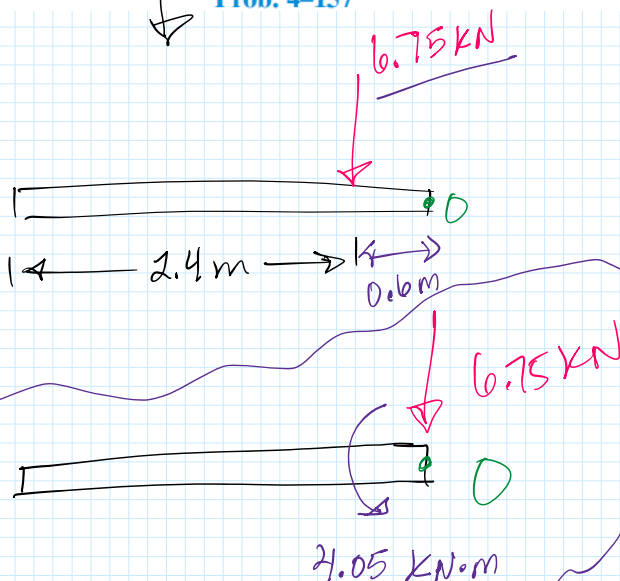
$$-109.9 = -133.3 \bar{x}$$

$$\bar{x} = 0.82 \text{ ft}$$

4-157. Determine the equivalent resultant force and couple moment at point O.



Prob. 4-157



$$F_R = \int_L w(x) dx \quad \bar{x} = \frac{\int_L x \cdot w(x) dx}{F_R}$$

$$F_R = \int_0^3 \frac{1}{3} x^3 dx = \frac{1}{12} x^4 \Big|_0^3$$

$$F_R = \frac{1}{12} (3)^4 - \frac{1}{12} (0)^4 = 6.75 \text{ kN}$$

$$F_R = 6.75 \text{ kN} \downarrow$$

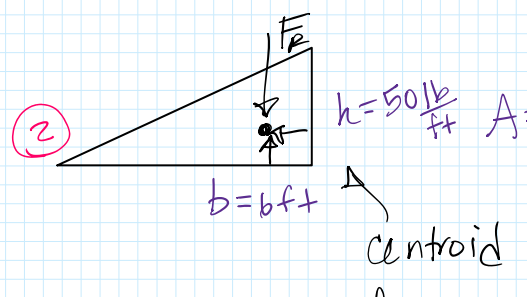
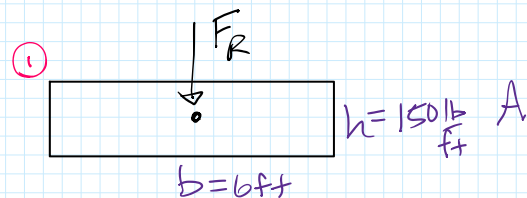
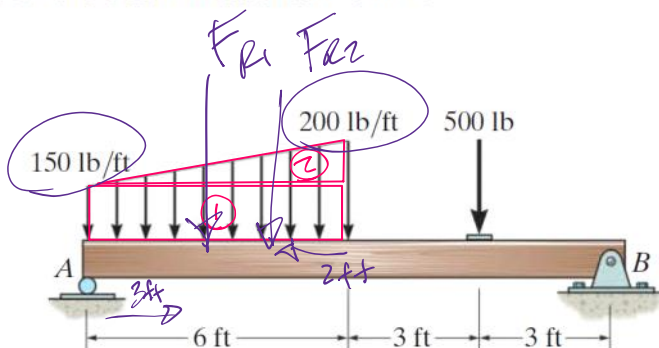
$$\bar{x} = \frac{\int_0^3 x \left(\frac{1}{3} x^3 \right) dx}{6.75} = \frac{\int_0^3 \frac{1}{3} x^4 dx}{6.75}$$

$$\bar{x} = \frac{\frac{1}{15} x^5 \Big|_0^3}{6.75} = \frac{\frac{1}{15} (3)^5}{6.75} = \frac{16.2}{6.75}$$

$$\bar{x} = 2.4 \text{ m}$$

$$M_O = 6.75 (0.6 \text{ m}) = 4.05 \text{ kN} \cdot \text{m}$$

F4-40. Determine the resultant force and specify where it acts on the beam measured from A.





$$= b \cdot h$$

$$= \frac{1}{2} \cdot b \cdot h$$

$$\text{is } \frac{1}{3}$$



Prob. F4-40

2-077

centroid
from either

$$F_{R1} = \left(150 \frac{\text{lb}}{\text{ft}}\right) (6 \text{ ft}) = 900 \text{ lbs}$$

$$F_{R2} = \frac{1}{2} (6 \text{ ft}) \left(50 \frac{\text{lb}}{\text{ft}}\right) = 150 \text{ lbs}$$

is $\frac{1}{3}$
ner base